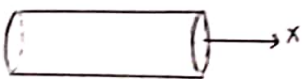
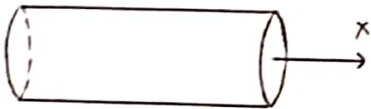
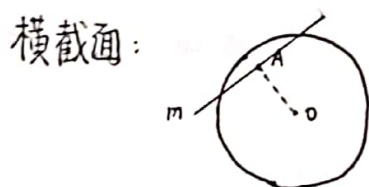


1. $\beta - \alpha$

2. 0.05 ; 0.637 GPa ; 0.25

3. 轴向拉力:  取与 x 轴平行的平面

扭矩:  以 A 点为例, 要取与 x 轴平行, 且与 OA 连线垂直的平面 m



4. (1) AC 杆上最大扭转剪应力 $\tau_{max} = \frac{M_e}{W_p} = \frac{16M_e}{\pi d^3} = 50.93 \text{ MPa} < 75 \text{ MPa}$
 AB 筒上最大扭转剪应力 $\tau_{max} = \frac{M_e}{W_p'} = \frac{16M_e}{\pi D^3(1-\alpha^4)} = 5.52 \text{ MPa} < 10 \text{ MPa}$

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(2) B 端固定不动, AB 扭转角度 $\alpha = \frac{M_e \cdot |AB|}{G_{AB} \cdot I_p'} = \frac{M_e \cdot |AB| \cdot 32}{G_{AB} \cdot \pi D^4(1-\alpha^4)} = 1.767 \times 10^{-3}$

AC 的扭转角度为 $\beta = \frac{M_e \cdot |AC|}{G_{AC} \cdot I_p} = \frac{M_e \cdot |AC| \cdot 32}{G_{AC} \cdot \pi d^4} = 0.122$

C 端转角为 $\alpha + \beta = 0.124$, 换算成角度为 7.1°



5. 在本题截面坐标为 (y, z) 处, 受到的应力大小为 $\sigma_{(y, z)}$

$$\sigma_{(y, z)} = -\frac{M_z y}{I_z} + \frac{M_y z}{I_y}, \quad \text{而 } I_z = \frac{40 \text{ mm} \times 80^3 \text{ mm}^3}{12} = 4I, \quad I_y = \frac{80 \text{ mm} \times 40^3 \text{ mm}^3}{12} = I$$

($I = 4.2667 \times 10^{-7} \text{ m}^4$)

(1) $M_y = 40 \text{ kN} \cdot \text{m}$, $M_z = 0$, 代入 $\sigma_{(y, z)} = 0$, 得

$$-\frac{M_z y}{I_z} + \frac{M_y z}{I_y} = 0, \quad z = 0, \quad \text{中性轴为 } z = 0$$

(2) $M_y = 40 \text{ kN} \cdot \text{m}$, $M_z = 80 \text{ kN} \cdot \text{m}$, 代入 $\sigma_{(y, z)} = 0$, 得

$$-\frac{80 \text{ kN} \cdot \text{m}}{4I} \cdot y + \frac{40 \text{ kN} \cdot \text{m}}{I} \cdot z = 0, \quad y - 2z = 0, \quad \text{中性轴为 } y - 2z = 0$$

